

VetMyPet - AI Animal Health & Outbreak Surveillance Platform

An End-to-End Multimodal Clinical AI, Geospatial Outbreak Sensing & Coordinated Epidemic Defense System

Problem Statement ID PS 26128 (Govt of Maharashtra)

Platform Core Go 1.22+ / Triple SQLite WAL

Clinical AI Engine viggovet/viggoVet-Reasoning-20B

Document Edition v3.2.0 Production Release

1. Problem Statement Alignment (PS ID 26128)

Official Problem Statement 26128 • Government of Maharashtra & MSIS

Title: Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues

Description: Livestock owners, field veterinarians, para-veterinary workers and government departments often lack a unified, realtime mechanism to identify emerging animal-health risks at the village, block and district levels. Disease symptoms may be reported late, diagnostic facilities may be distant, vaccination and treatment histories may be incomplete, and information from farms, veterinary dispensaries, laboratories, vaccination drives and surveillance programmes may remain fragmented. These gaps can delay containment, increase livestock mortality and productivity loss, raise the risk of zoonotic transmission, and affect farmers' incomes. The challenge is to create a practical system that enables early warning, rapid reporting, risk assessment, preventive action, referral and coordinated response.

The **VetMyPet** platform is engineered as a direct solution to this mandate, uniting dairy farmers, pastoralists, licensed veterinary doctors, dispensaries, NGOs, and state authorities into an interconnected, real-time epidemiological grid.

2. Multi-Role Ecosystem & Geospatial Entity Mapping

To address fragmentation between field workers and clinical facilities, the platform partitions functionality across four primary user classes:

 Livestock Farmers & Dairy Owners (farmer)

Registers farm name, village (गावाचे नाव), taluka, district, livestock types (Cattle, Buffalo, Goat, Sheep, Poultry, Swine), and herd size. Features livestock AI video/photo screening, real-time outbreak SOS radar, recovery reporting (POST /api/outbreaks/report-recovery), and nearby clinic directory.

 Companion Pet Parents (pet_owner)

Registers address, city, district, and pincode. Features instant AI lesion & motion gait scanner, broad-spectrum contagion alerts (e.g. Canine Parvovirus, Feline Panleukopenia), and one-tap doctor second-opinion requests.

 Licensed Veterinarians & Clinics (vet)

Dedicated Doctor Decision Hub: Case Review Queue to inspect doubtful AI scans, verify diagnoses, issue Rx prescriptions, manage live clinic operating hours, broadcast current field visiting locations, and post leave notices.

 NGOs & Government Dispensaries (ngo / gov)

Real-time emergency inventory management for critical vaccines (FMD, Lumpy Skin, PPR), broad-spectrum antibiotics, antiseptics, and PPE kits with automated low-stock warnings and official circular broadcasting.

3. Dynamic Outbreak Sensing, Vet Consensus & Resolution Engine

Unlike static alert placeholders, VetMyPet incorporates a dynamic epidemiological state engine:

 Rule 1: Veterinarian Consensus Trigger

When 3 licensed veterinarians confirm the same infectious disease (e.g. FMD, Lumpy Skin, PPR, Parvo, Anthrax) within a

district, the system automatically flags a **Verified Outbreak** (`status = 'active'`, `severity = 'CRITICAL'`) and dispatches

district-wide SOS alerts.

Rule 2: Farmer Clustering Trigger

When 3 farmers in the same locality/taluka report matching clinical symptoms via multimodal scans or consult requests, an

active outbreak cluster is automatically formed with actionable biosecurity protocols.

Dynamic Outbreak Resolution Lifecycle

Recovery Reporting: Farmers or paravets submit recovery counts (`POST /api/outbreaks/report-recovery`). Active affected counts decrement dynamically.

Containment Clearance: When affected counts reach 0 or when veterinarians issue an official resolution certificate (`POST /api/outbreaks/resolve`), the outbreak transitions to `status = 'resolved'`, `severity = 'RESOLVED'`, the active red SOS flag is cleared, and an all-clear resolution notice is broadcast across the district.

4. End-to-End Architecture & Data Flow Diagram

```
graph TD
    subgraph Layer1 [ ]
        direction LR
        L1_1[VETMYPET FRONTEND SPA (React 18 / Tailwind)]
        L1_2["[Farmers: Herd Scanner & Outbreak SOS]"]
    end

    subgraph Layer2 [ ]
        direction LR
        L2_1["[Vets: Case Queue & Duty Management]"]
        L2_2["[Directory & Biosecurity Defense Radar]"]
        L2_3["REST API (X-Auth-Token, X-User-ID, Multipart / JSON)"]
        L2_4["http://localhost:8080"]
    end

    subgraph Layer3 [ ]
        direction LR
        L3_1["GO HIGH-PERFORMANCE BACKEND (Go 1.22+)"]
        L3_2["Authentication Middleware & Bcrypt Security - Outbreak Cluster Sensing & Consensus Engine"]
        L3_3["FFmpeg Temporal Keyframe Extractor (4 Frames)"]
        L3_4["Asynchronous Backup Snapshot Hook (VACUUM INTO)"]
        L3_5["Multipart"]
    end

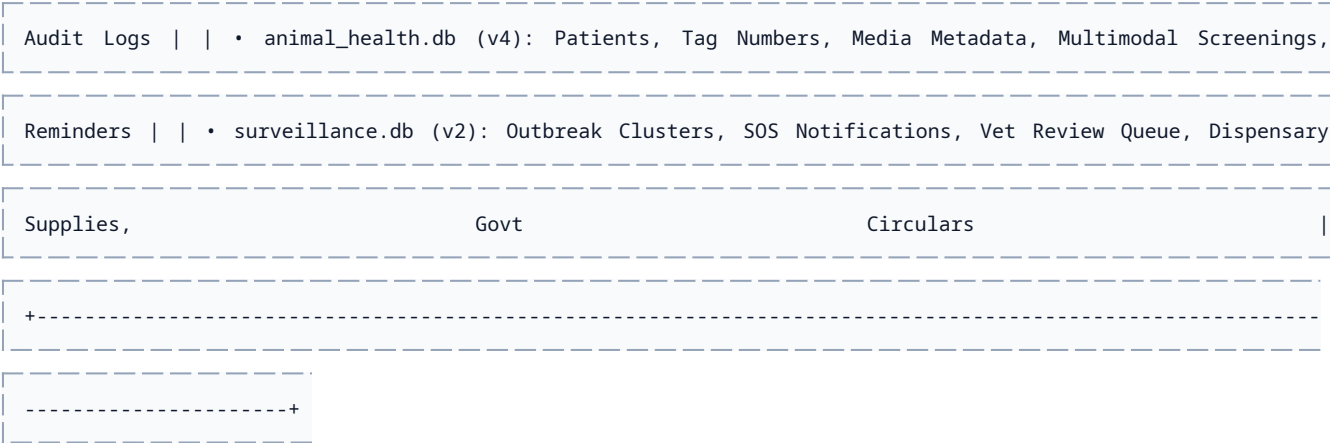
    subgraph Layer4 [ ]
        direction LR
        L4_1["Media Ingestion | AI Inference Request | Surveillance & Case Escalation"]
    end

    subgraph Layer5 [ ]
        direction LR
        L5_1["STORAGE & MEDIA ENGINE | DUAL-STAGE MULTIMODAL AI PIPELINE | CONSENSUS & RECOVERY | Content-Addressable Storage (./uploads/) | Stage 1: Qwen2.5-VL-72B-Instruct (Vision) | Vet Consensus ≥3 | FFmpeg 4-Frame Keyframe Generator (JPEG) | Stage 2: viggovet/viggoVet-Reasoning-20B"]
    end

    subgraph Layer6 [ ]
        direction LR
        L6_1["Farmer Cluster ≥3"]
        L6_2["Persistence across Segregated Databases"]
    end

    subgraph Layer7 [ ]
        direction LR
        L7_1["TRIPLE-DATABASE ARCHITECTURE (SQLite WAL) | user.db (v4): Profiles, Roles, Geolocation, Clinic Duty & Leave Notices, Bcrypt Credentials"]
    end

    Layer1 --- Layer2
    Layer2 --- Layer3
    Layer3 --- Layer4
    Layer4 --- Layer5
    Layer5 --- Layer6
    Layer6 --- Layer7
```



5. Triple-Database Architecture & Migration Specifications

To ensure absolute data isolation, zero cross-table lock contention, and audit compliance, the system maintains three segregated databases:

Database	Table Entity	Schema Columns & Role Definition
user.db Schema (WAL)	users	id (PK), email (UNIQUE), role, password_hash, phone, address, city, district, state, pincode, farm_name, farm_village, farm_taluka, livestock_types, herd_size, clinic_name, clinic_address, clinic_hours, clinic_availability, clinic_visiting_location, unavailability_notice, created_at, updated_at
	sessions	token (PK), user_id (FK), user_agent, ip_address, expires_at, created_at
	auth_logs	id (PK), user_id, email, event, ip_address, details, created_at
	animals	id (PK), user_id (INDEX), name, species, breed, tag_number, herd_size, village, taluka, district, notes, created_at
animal_health.db Schema (WAL)	media	id (PK), url, path, type, mime_type, size, created_at
	health_screenings	id (PK), animal_id, user_id, media_url, symptoms_json, visual_analysis_json, assessment_json, urgency, district, taluka, farm_name, created_at
	reminders	id (PK), user_id, animal_id, type, title, description, due_at, completed, created_at
surveillance.db Schema (WAL)	outbreaks	id (PK), disease_name, species, district, taluka, village, farm_name, affected_count, severity, prevention_guide, status ('active' 'contained' 'resolved'), verified_by_vets, reported_by_farmers, resolution_notes, resolved_at, created_at
	notifications	id (PK), user_id, role_target ('all' 'farmer' 'vet' 'ngo'), district, title,

Database	Table Entity	Schema Columns & Role Definition
		message, outbreak_id, severity, is_sos, read, action_url, created_at
	vet_consultations	id (PK), screening_id, user_id, user_name, user_role, user_phone, location, vet_id, vet_name, animal_name, species, breed, media_url, symptoms_json, owner_notes, ai_diagnosis, ai_urgency, vet_diagnosis, vet_suggestion, vet_prescriptions, status ('pending' 'reviewed'), created_at
	inventories	id (PK), user_id, owner_name, org_type, district, item_name, category, quantity, unit, min_threshold, expiry_date, status, updated_at
	gov_advisories	id (PK), issuer, title, content, district, category, urgency, date_issued, created_at

6. Complete RESTful API Specifications

Method & Path	Headers & Params	Description & Payload Architecture
POST /api/auth/register	Content-Type: json	Registers account with role (farmer, pet_owner, vet, ngo) and geo fields (farm name, village, taluka, clinic hours).
POST /api/auth/login	Content-Type: json	Authenticates credentials with Bcrypt verification; issues 30-day session token.
POST /api/auth/profile	X-Auth-Token	Updates profile, farm parameters, or veterinarian availability duty status & leave notices.
POST /api/auth/change-password	X-Auth-Token	Changes user password with old password verification and audit logging.
GET /api/vets	?district=Pune	Returns directory of licensed veterinarians with real-time duty status, visiting location, and clinic hours.
POST /api/health-check/upload	multipart/form-data	Validates and writes image/video to content-addressable storage.
POST /api/health-check/analyze	X-Auth-Token, X-User-ID	Executes 4-frame temporal extraction, visual AI inspection, and veterinary medical differential assessment.
GET /api/outbreaks	?district=...&species=...	Lists active and resolved disease clusters across Maharashtra districts.
GET /api/outbreaks/:id	Public / Auth	Fetches complete outbreak record with consensus stats, affected counts, and biosecurity guidelines.
POST	Content-Type: json	Reports herd recovery. Automatically decrements affected

Method & Path	Headers & Params	Description & Payload Architecture
POST /api/outbreaks/report-recovery		counts and clears SOS when resolved.
POST /api/outbreaks/resolve	X-Auth-Token (Vet/Gov)	Issues official veterinary clearance certificate and containment notice.
GET /api/notifications	X-Auth-Token	Fetches targeted and broadcast SOS alerts and biosecurity notices.
POST /api/vet-consultations	X-Auth-Token	Farmers/Pet Owners request second opinion on doubtful AI scans.
POST /api/vet-consultations/review	X-Auth-Token (Vet)	Doctor submits clinical diagnosis, advice, and Rx prescriptions. Triggers consensus check.
GET /api/inventory	?district=...	Returns dynamic stock of vaccines, antibiotics, antiseptics, and PPE supplies.
GET /api/gov-advisories	?district=...	Returns official Maharashtra animal husbandry circulars and ring vaccination directives.

7. Clinical Verification Benchmarks across Livestock & Companion Animals

Clinical Scenario	Visual Analysis (Qwen-VL)	Veterinary Differentials (ViggoVet)	Triage & Outbreak Status
Bovine Cluster Baramati Farm, Cattle fever & salivation	Farm Excessive ropy salivation, interdigital foot vesicles, oral erosions on tongue, lameness.	1. [HIGH] Foot-and-Mouth Disease (FMD / लाळ्या खुरकूत) 2. [MODERATE] Vesicular Stomatitis 3. [POSSIBLE] Bovine Viral Diarrhea (BVD)	CRITICAL SOS Verified Outbreak Active
Caprine Sangamner Herd, fever & erosions	Epidemic Goat Mucopurulent discharge, necrotic dyspnea, diarrhea	1. [HIGH] Peste des Petits Ruminants (PPR) 2. [MODERATE] Contagious Caprine Pleuropneumonia 3. [POSSIBLE] Goat Pox	URGENT Vaccine Deployed
Canine 10s video of puppy twitching	Motion Clip chewing-gum seizures, hyperkeratosis.	1. [HIGH] Canine Distemper Encephalomyelitis 2. [MODERATE] Idiopathic Epilepsy	EMERGENCY Doctor Consult Sent

Clinical Scenario	Visual Analysis (Qwen-VL)	Veterinary Differentials (ViggoVet)	Triage & Outbreak Status
Healed Dairy Herd Malegaon Budruk Day 14 Follow-up	Complete re-epithelialization of oral lesions, normal gait, voluntary feeding resumed.	3. [POSSIBLE] Viral Meningoencephalitis	
		1. [RESOLVED] Post-FMD Clinical Recovery	RESOLVED
		2. Complete tissue healing confirmed.	SOS Cleared • Movement Open